Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Phase 1: Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

1. Team & Project Overview

Team Name	AI NewBee
Selected Domain	Domain 1 – Aiversecity Future Ready Workforce
Selected SDG Track (1–17)	SDG 4 – Quality Education
Project / Solution Name	Minder
Name (Leader)	Lee Yi Cheng
Email & Phone (Leader)	yicheng070612@gmail.com / +601118903388
University Affiliation(s)	Asia Pacific University / Universiti Teknologi Malaysia
Number of Team Members	3

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

When technology impacts the skills of various occupations, early career and students are left with uncertainty. They can search for jobs, courses and AI trends but still have a hard time evaluating which jobs would benefit from their current skills, what skills they are missing and what steps to take next. It can be challenging to judge recommendations that are not supported by reasons or dated evidence. If a plan reaches a list of courses, the learner has little means to pursue the plan.

Our target user is a Malaysian university student, graduate or career changer who is interested in exploring opportunities in digital and AI. The problem we are addressing in the real world is from information about careers to an individual learning decision. The objective of Minder is to link a person's interests, self-reported skills, aims and time constraints to the role options that can be explained, the skills gaps and the learning tasks that can be achieved. This suggested pathway is relevant to skills development and the initial prototype does not show an improvement in hiring or wages.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Minder is a proposed career exploration and reskilling application that leverages AI. At the heart of its core workflow, a learner can build a profile of goals, interests, skills, experience and learning-time constraints. A career matching layer compares the roles that are supported to a list of options and returns a list of ranked options, along with explanations, skill gaps, and a clearly defined scoring mechanism. A compatibility score is not a job guarantee, but an index that helps make a decision.

The Explore area is a cluster of experiences based on the same user data. Career Match compares jobs; Skills shows job skills and skills gaps; Tasks shows steps to a chosen pathway; Progress shows task completion as recorded by the learner; Future Explorer compares dated market observations and alternate scenarios with assumed observations where evidence is available. SDG links are used to make learning activities relevant to learning skills goals.

The planned implementation uses React and TypeScript for the interface, a Python/FastAPI application boundary for persistence and orchestration, and a separate matching interface with a transparent baseline or approved experimental model. Supporting layers are offline modelling and an optional chat assistant based

on Groq which has bounded agents. The prototype for the frontend has been done with Replit. The team plans to use the GenAI API integration that was developed in Phase 1 for the assistant, the database architecture developed in Phase 1 for the persistence layer, and cloud deployment of the prototype developed in Phase 1. In Phase 2, the application features will be clearly separated from any technical experiments.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Web interface: React + TypeScript + Vite, built in a Replit workspace.

Application API: Python, FastAPI and Pydantic for versioned routes, input validation and orchestration

Persistence: PostgreSQL target with SQLAlchemy and Alembic for profiles, skills, role catalogues, plans, tasks, progress and versioned recommendation runs.

Matching and evaluation: Python deterministic baseline, with Ridge/XGBoost experiments under evaluation. The backend owns all scoring; no hiring-probability or validated-demand claims are made.

Market evidence and scenarios: Dated, source-tagged, Malaysia-first records with a role/skill taxonomy and a conditional scenario pipeline; current observations and future assumptions are kept separate.

LLM and agents (optional): Groq-hosted assistant (MindBot) with a bounded tool layer that explains saved results and asks for user approval before changing plans; counted as a deployed feature only once verified in the user interface.

Tooling and offline compute: GitHub for source control and Replit for frontend work. RunPod for offline training

System block diagram (data flow):

(Stage 1) The React/TypeScript app collects profile changes, career selection and task actions

(Stage 2) FastAPI validates the session and calls shared profile, career, pathway and progress services

(Stage 3) A relational database holds profiles, role requirements, plans, tasks, trend provenance and recommendation versions

(Stage 4) The matching and scenario engine scores profile–role pairs and explains gaps using dated evidence

(Stage 5) The optional Groq assistant explains saved state and requests approval for changes

(Stage 6) Model training and evaluation run offline; only a fixed approved artifact or deterministic matcher serves requests.

YES. Our project utilized official SDC sponsor-provided platform credits and accounts (including Replit cloud platform and developer environment), which served as our primary workspace for rapid frontend prototyping, application orchestration, and development integration

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We framed the user problem around a specific decision: after seeing possible future roles, can a learner understand the reason for a match and take a manageable next step? The first user journey uses a fictional student profile and moves through role comparison, skill gaps, a selected learning pathway, tasks and progress. This reduces interface complexity and keeps the outcome testable.

Development is organised as a React/TypeScript interface connected to a versioned API, persistence and a separate matching component. We prioritise input validation, isolated demo sessions, repeatable seeded data, clear loading and error states, and task persistence after refresh. The matching interface can use a transparent deterministic baseline until a trained model is supported by reviewed labels and proper evaluation. Evidence and future scenarios must disclose dates, assumptions and coverage limits.

Before Phase 2, the team will verify the actual end-to-end demo, document reproducible setup, run a small usability or task-timing benchmark and correct the most important usability failures. User interviews, scale testing and model generalization are not claimed unless measured.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Primary target 4.4: increasing relevant skills for employment, decent jobs and entrepreneurship. UN source https://sdgs.un.org/goals/goal4.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☐ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☐ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☐ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React	Code	MIT
TypeScript	Code	Apache-2.0
Vite	Code	MIT
FastAPI	Code	MIT
Pydantic	Code	MIT
SQLAlchemy	Code	MIT
Alembic	Code	MIT
PostgreSQL	Code(database)	PostgreSQL License

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team’s knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]

[Date]